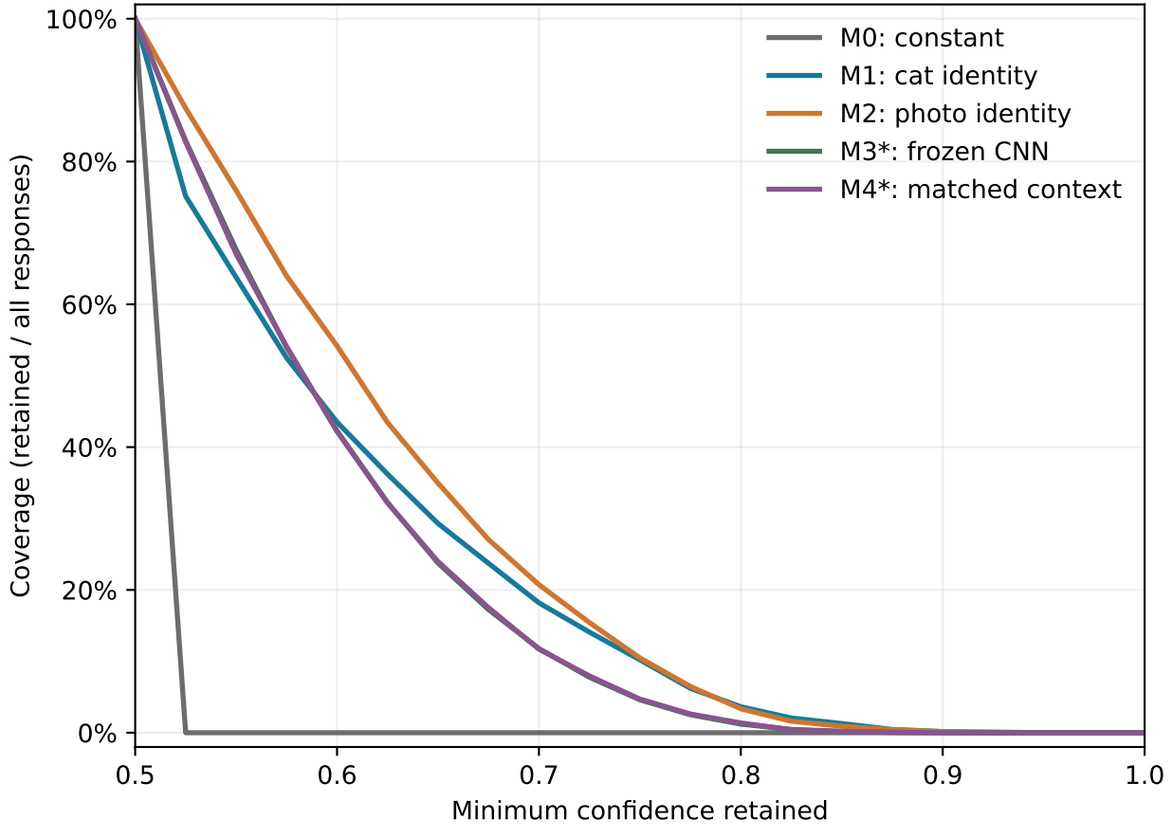
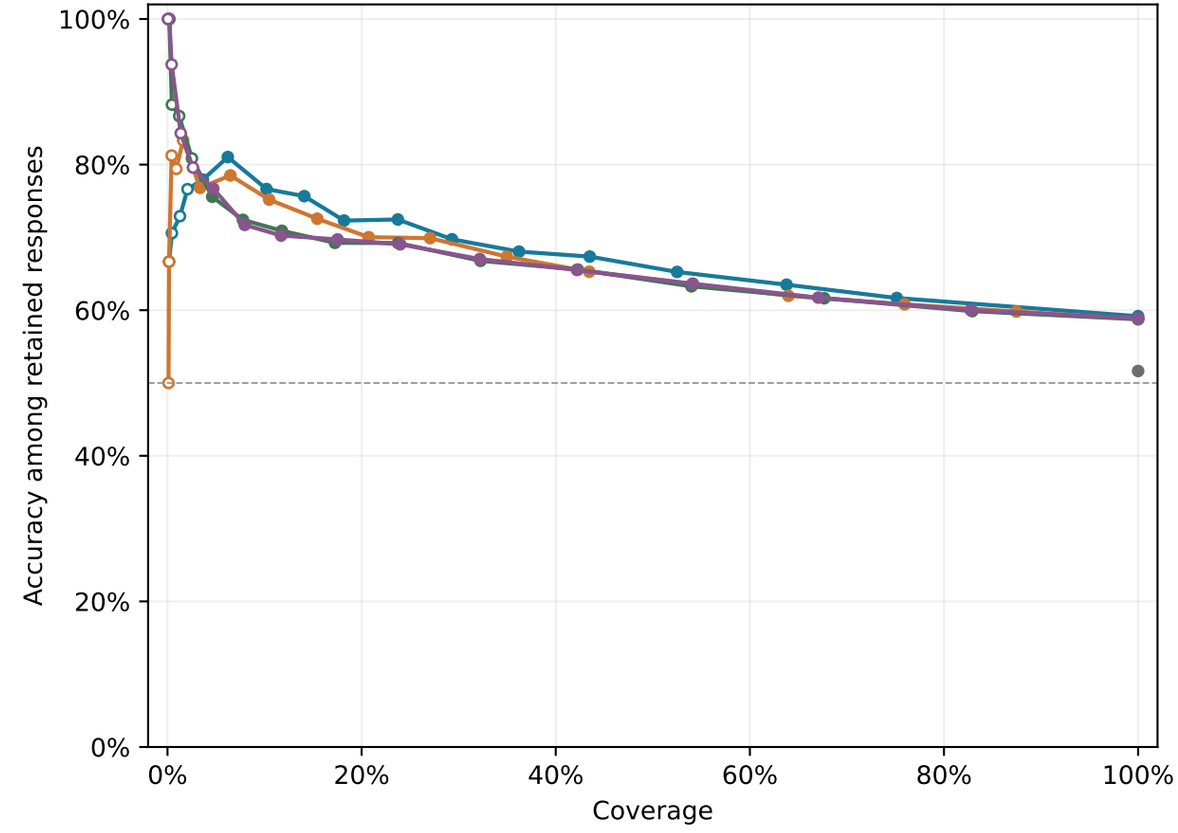
Abstention: confidence, coverage, and selective accuracy

Post-Test Exploratory Analysis | Not Threshold Selection





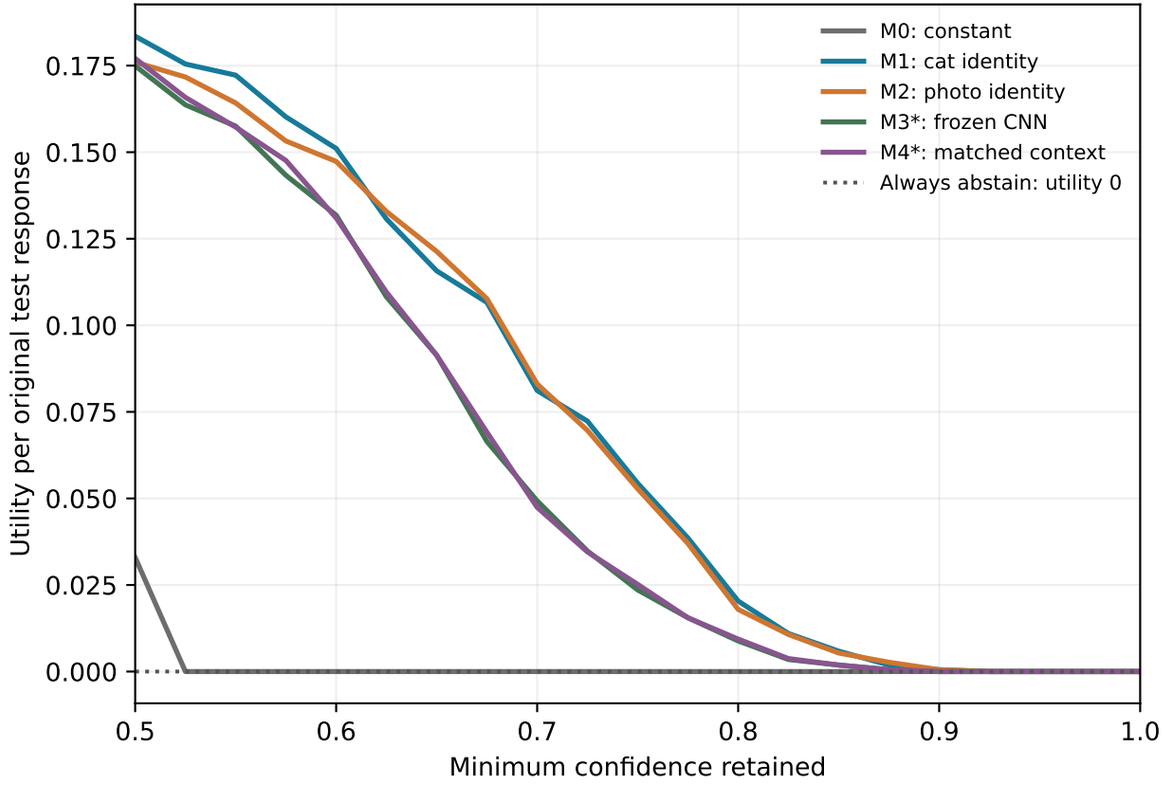
Confidence = max(P(A), 1-P(A)). Retain confidence >= threshold; tied confidence values stay together.

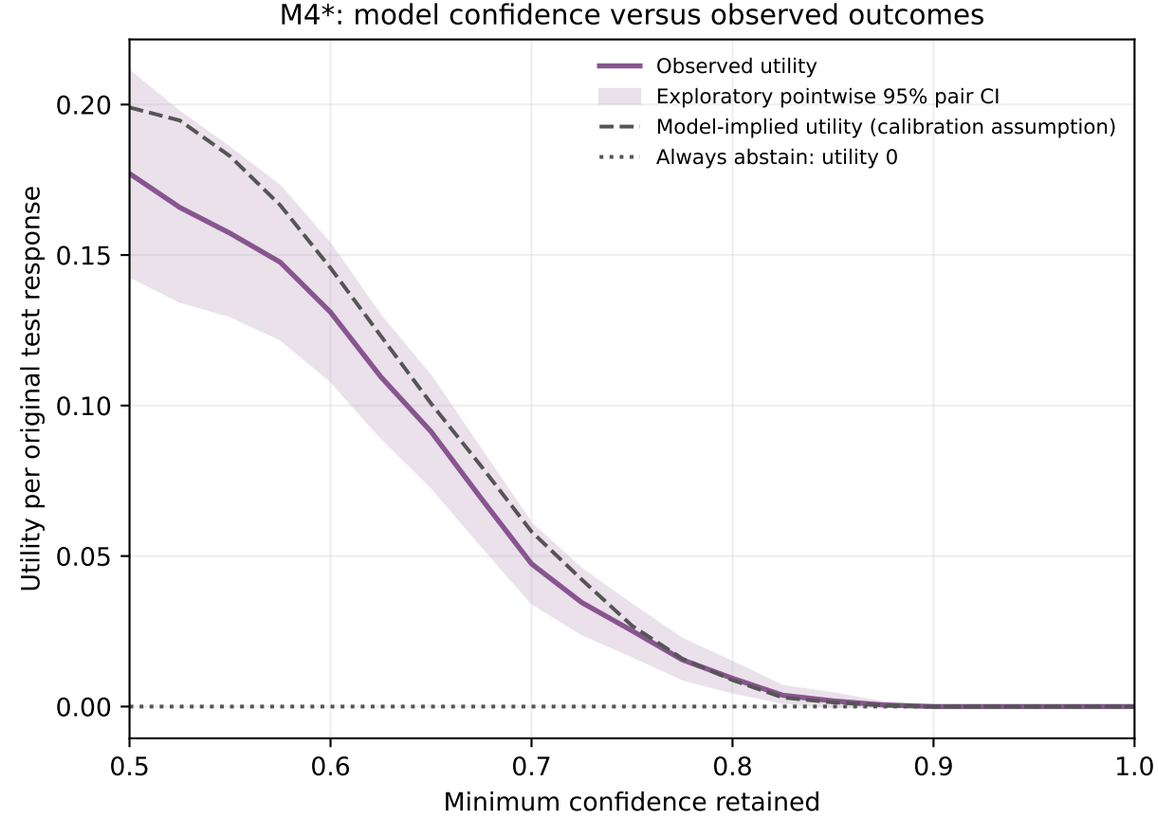
Filled points: at least 100 retained responses and 30 pairs. Hollow points: low support; use caution.

Different models retain different rows. Empty selections have undefined accuracy, not 0% or 100%.

Utility: correct +1, incorrect -1, abstention 0

Post-Test Exploratory Analysis | Not Threshold Selection





Utility = (correct retained - incorrect retained) / all original responses, not just retained cases.

With these payoffs and calibrated probabilities, always predict is weakly optimal in expected total utility.

Higher selective accuracy alone is not a utility gain. Model-implied utility is not guaranteed real-world utility.

Thresholds are fixed for display; no test-optimal threshold or deployment policy is selected.

Fixed illustrative thresholds: counts and utility

Post-Test Exploratory Analysis | Not Threshold Selection

Model	Threshold	Retained N	Retained pairs	Coverage	Retained accuracy	Utility / all N	Support
M0	0.50	3733	1207	100.0%	51.6%	+0.0329	supported
M0	0.60	0	0	0.0%	N/A	+0.0000	no retained cases
M0	0.70	0	0	0.0%	N/A	+0.0000	no retained cases
M1	0.50	3733	1207	100.0%	59.2%	+0.1835	supported
M1	0.60	1624	588	43.5%	67.4%	+0.1511	supported
M1	0.70	679	242	18.2%	72.3%	+0.0812	supported
M2	0.50	3733	1207	100.0%	58.8%	+0.1760	supported
M2	0.60	2022	746	54.2%	63.6%	+0.1473	supported
M2	0.70	774	291	20.7%	70.0%	+0.0830	supported
M3*	0.50	3733	1207	100.0%	58.7%	+0.1749	supported
M3*	0.60	1578	596	42.3%	65.6%	+0.1318	supported
M3*	0.70	440	172	11.8%	70.9%	+0.0493	supported
M4*	0.50	3733	1207	100.0%	58.9%	+0.1771	supported
M4*	0.60	1575	619	42.2%	65.5%	+0.1310	supported
M4*	0.70	437	180	11.7%	70.3%	+0.0474	supported

Common frozen population: 3,733 responses, 1,207 unordered pairs. Values are descriptive.

The complete fixed grid, pointwise intervals, and undefined-resample counts are saved in the accompanying CSV.

Low-support rows are displayed transparently; do not interpret them as reliable deployment evidence.

Interpretation, safeguards, and limitations

Post-Test Exploratory Analysis | Not Threshold Selection

- Data: saved frozen common strict-G2 predictions only. No raw withheld choices are loaded; no refit or new model selection is performed.

- These are illustrative payoffs, not measured error costs or a claim that human decisions should use this utility.

- Selective accuracy conditions on accepted responses. Utility includes every original response, with abstentions contributing zero.

- A constant-confidence baseline retains all or none at a cutoff. No arbitrary tie-breaking or exact-coverage quota is used.

- Pair bootstrap keeps all responses and sampled-pair multiplicity. Conditional accuracy intervals require adequate retained support and no empty bootstrap selections.

- Intervals are exploratory and pointwise, not multiplicity-adjusted. They omit training variability, respondent linkage, shared-image dependence, and prior analysis-induced selection.

- No optimum is picked from the test curves. A deployment threshold requires explicit decision costs and an independent validation policy study.